

Law of Falling Bodies

Gravity pulls more strongly on heavier objects—but heavier objects need more force to make them speed up than lighter ones. Galileo was the first person to realize, in 1590, that any two objects dropped together should speed up at the same rate and hit the ground together. We are used to lighter objects falling more slowly—because air resistance slows them more.



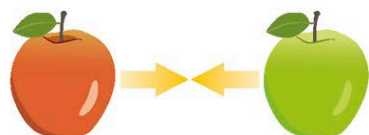
In the near-frictionless environment of the moon, a heavy hammer and a light feather fall at the same rate.

Falling in a vacuum

In 1971, astronaut Dave Scott proved Galileo right when he dropped a feather and a hammer on the moon.

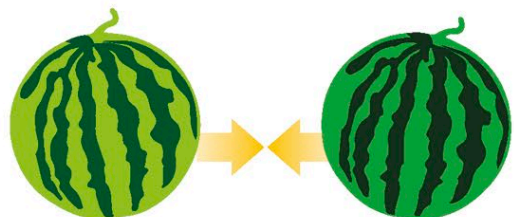
Law of Universal Gravitation

In 1687, English scientist Isaac Newton came up with his Law of Universal Gravitation. It states that any two objects attract each other with a force that depends on the masses of the objects and the distance between them.



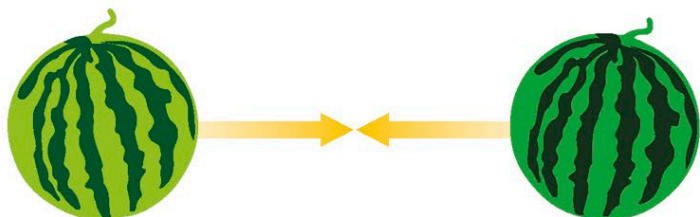
Equal and opposite

The gravitational force between two objects pulls equally on both of them—whatever their relative mass—but in opposite directions.



Double the mass

If one object's mass is doubled, the gravitational force doubles. If the mass of both objects is doubled (as here), the force is four times as strong.



Double the distance

If the distance between two objects is doubled, the gravitational force is quartered.

Gravity and orbits

Newton used his understanding of gravity (see left) and motion to work out how planets, including Earth, remain in their orbits around the sun. He realized that without gravity Earth would travel in a straight line through space. The force of gravity pulls Earth toward the sun, keeping it in its orbit. Earth is constantly falling toward the sun, but never gets any closer. If Earth slowed down or stopped moving, it would fall into the sun!

Elliptical orbit

Earth's orbit around the sun is in fact elliptical (an oval), not circular.

Speed of travel

If Earth was not speeding through space, gravity would pull it into the sun.



Earth

Based on the strength of its gravitational force, the mass of Earth is estimated to be 6.5 sextillion tons!

The force of gravity 62 miles (100 km) above Earth is **3 percent less** than at sea level on Earth.